Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 20 epochs with a batch size of 64. The learning rate was set to 0.001 with the Adam optimizer.

2 Results

Figure 1 shows the f1 score by dataset.

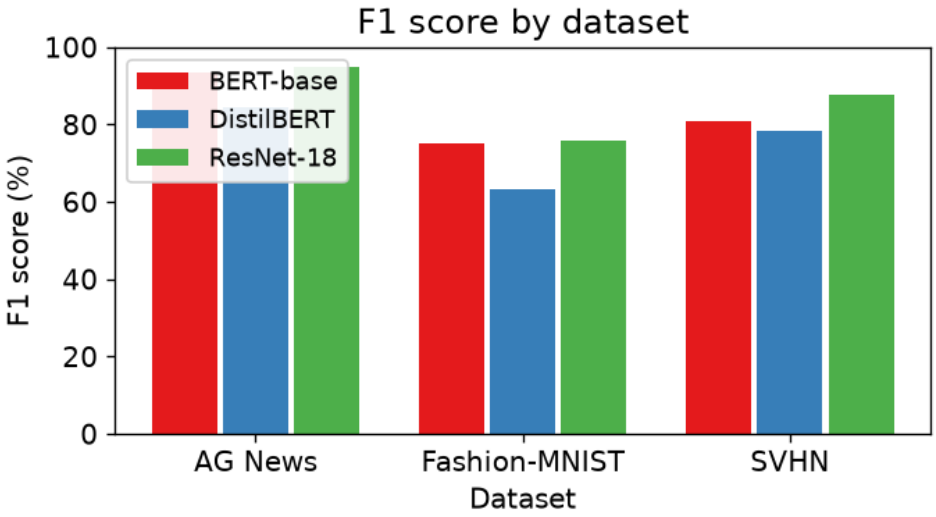


Figure 1: F1 score by dataset for the models in this experiment.

As shown in Figure 1, on AG News, ResNet-18 is 11.6 points better than BERT-base. ResNet-18 reaches 76% F1 score on Fashion-MNIST (Figure 1). ResNet-18 reaches 76% F1 score on Fashion-MNIST (Figure 1). On SVHN, ResNet-18 is 9 points better than DistilBERT (Figure 1).

Figure 2 shows the validation accuracy during training.

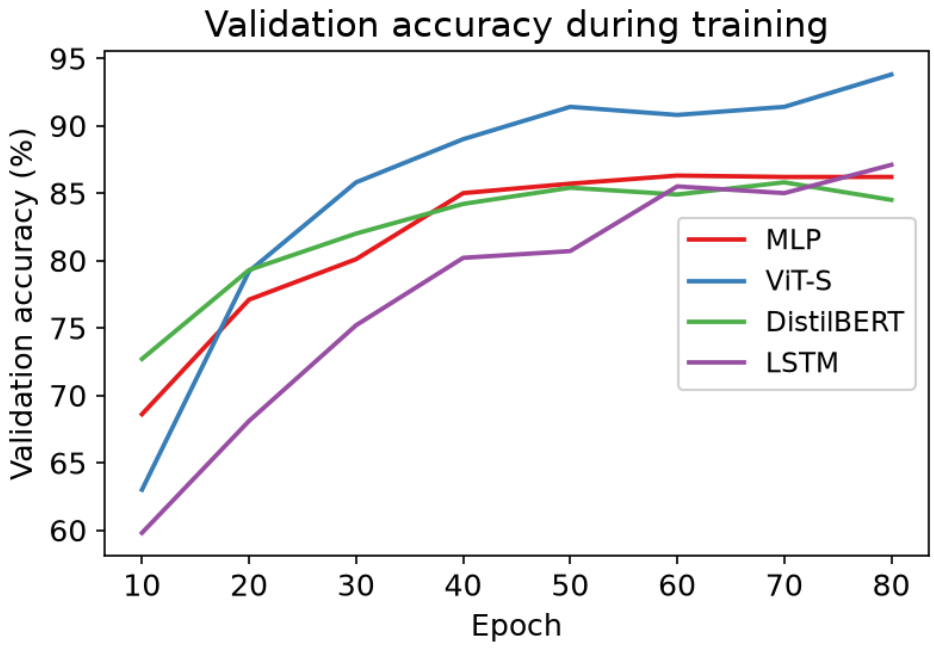


Figure 2: Validation accuracy during training for the models in this experiment.

DistilBERT reaches 86.5% validation accuracy at epoch 70. As shown in Figure 2, MLP reaches 86% validation accuracy at epoch 60.

Figure 3 shows the f1 score during training.

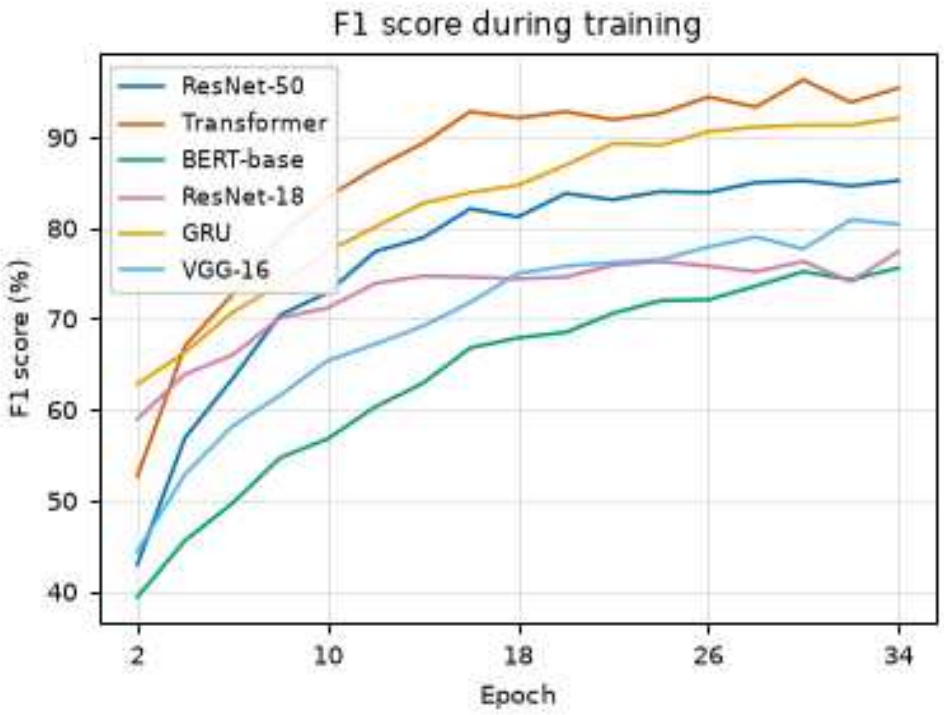


Figure 3: F1 score during training for the models in this experiment.

From epoch 30 to epoch 34, the F1 score of GRU improves by a relative 0.9% (Figure 3). ResNet-18 reaches 76.3% F1 score at epoch 24. As shown in Figure 3, at epoch 24, ResNet-50 is 7.5 points better than VGG-16.

3 Discussion

On MNIST, XGBoost obtains an accuracy of 59.5%. We note that the training was stable for all models.